

Moolya QA Services and Solutions

Comprehensive Quality Engineering and Testing Services designed to help product and engineering teams build reliable, high-performing, and user-centric applications.



Flexible Engagement Models

Our engagement models range from full QA ownership to staff augmentation, enabling clients to scale quality operations with speed, efficiency, and measurable outcomes.

Full Ownership

End-to-end QA governance with complete accountability

Co-Ownership

Shared responsibility between Moolya and client QA teams

Staff Augmentation

Skilled QA professionals integrate seamlessly into your ongoing projects



QA Ownership Model

When quality assurance needs to deliver tangible outcomes, Moolya takes full ownership of the QA process. We define, execute, and continuously improve test strategies to ensure product reliability, user satisfaction, and faster go-to-market.



Functional Testing

End-to-end test ownership across functional, regression, exploratory, and usability areas



Automation Testing

Scalable automation frameworks with CI/CD integration for faster feedback



Performance Testing

Load, stress, and scalability validation for server and client-side performance



Security Testing

Comprehensive VAPT aligned to OWASP Top 10 and compliance frameworks



Functional Testing Ownership

Challenges We Address

- Testing handled by developers with limited QA focus
- QA bandwidth consumed by release testing, leaving regression untested
- High production defect rate despite internal QA
- Pre-launch app readiness validation required
- Crowd-sourced testing for real-world conditions
- Inefficient QA processes lacking structure
- Need for usability and UX validation

Moolya's Approach

Risk-Based Testing

Context-driven and Growth-driven testing aligned to product goals

Dedicated QA Pods

Agile alignment with client delivery teams

Proactive Quality Monitoring

Continuous visibility into defects, trends, metrics, and release readiness

❏ **Outcome:** Predictable quality, reduced defect leakage, and improved product usability



Automation Testing Ownership

Challenges We Address

- Lengthy manual regression cycles
- Limited or no automation framework
- Growing automation backlog without governance
- Lack of in-sprint automation and CI/CD integration
- API and backend automation gaps
- Lack of a scalable automation foundation
- Unstructured and inefficient existing automation

Moolya's Approach

Framework Design

Modular, scalable automation frameworks

Regression Automation

Accelerate recurring release validation

CI/CD Integration

Faster feedback loops

Full-Stack Coverage

API, UI, backend, database

Tools/Frameworks: Playwright, Cypress, Selenium, Appium, WebdriverIO, **Robot Framework**

Languages: Java, JavaScript, Python, TypeScript, **C Sharp**

BDD Tools: Cucumber, Behave, SpecFlow

API Automation: Postman, REST Assured, Karate, **Playwright, Requests library**

📄 **Outcome:** Scalable automation architecture, Reduction in regression cycle time and improved release confidence



Performance Testing

Performance & Load Testing

Comprehensive validation for both server-side and client-side performance to ensure reliability and optimal user experience under real-world conditions.

Server-Side Testing

- Load, Stress, Endurance testing
- Scalability validation
- CPU and Memory Utilization
- Micro Services Capacity Testing
- Kafka Message Capacity Testing
- Webhook Performance Testing

Client-Side Testing

- Real-device performance analysis
- Page load time and rendering
- Front-end resource profiling
- Response time measurement
- UX impact under varying networks

❏ **Outcome:** Enhanced reliability, optimized performance, and improved scalability



Security & VAPT Testing

Comprehensive security assessments for web, mobile, and APIs aligned to OWASP Top 10 with compliance validation for data protection and regulatory frameworks.

- Web & Mobile Application Vulnerabilities
- API Security
- Network & Infrastructure Security
- Cloud & Configuration Security
- Penetration Testing

 **Outcome:** Strengthened application security posture and compliance assurance



Staff Augmentation

Moolya augments client teams with skilled QA professionals who bring proven experience across domains and technologies, ensuring immediate value and delivery continuity.

Engagement Model

01

Requirements Sharing

Clients share project scope and job description requirements

02

Testers Deployment

Moolya identifies and deploys the right-fit Testers

03

Seamless Integration

Testers align with client workflows, tools, and methodologies

Core Capabilities

- Manual & functional testing
- Automation scripting and maintenance
- Performance, API, and security testing
- Test consultancy and QA process setup

Value Proposition

- Rapid ramp-up of QA capacity with vetted experts.
- Flexible engagement models (onsite/offshore/hybrid).
- Domain-aligned testers (Healthcare, PharmaTech, SaaS, Life Sciences).
- Transparent rate card and governance.

❏ **Outcome:** Rapid scaling of QA capacity with domain-aligned and project-ready engineers



Moolya's QA Maturity Framework

Transforming industry challenges into competitive advantages through comprehensive quality engineering solutions.

QA Dimension	Industry Challenge	Moolya's Value Proposition
Functional Testing	High defect leakage	Context-driven, risk-based testing
Automation	Slow release cycles	Scalable frameworks and CI/CD integration
Performance	Bottlenecks discovered late	Continuous performance benchmarking
Security	Vulnerability exposure	Proactive VAPT and compliance checks
Staffing	Lack of Specialized QA Skills	Rapid ramp-up of QA capacity with vetted experts



Moolya's Solution-led Delivery

360° Testing Solution

Holistic testing for comprehensive coverage and risk mitigation, enhancing quality and satisfaction

Fastrack Automation

Dynamic, tool-agnostic automation to reduce time- to-market and ensure thorough test coverage

Rapid Deployment Testing Solution

Streamlined testing for faster release cycles and confident, quality product delivery

SpeedMaster Optimization Solution

Performance optimization to boost application speed, responsiveness, and scalability

Bugasura

Bug management tool, that helps you Report, track, and resolve issues efficiently with AI-enabled bug reporting and issue tracking

Secure Shield

Comprehensive security testing solution designed to protect your digital assets from vulnerabilities and threats

SREngine

Advanced solution for Service Reliability Engineering (SRE), ensuring highest levels of reliability, availability, and performance

MoolyaWave

DevOps services tailored to the unique needs of organizations developing web and mobile applications



Case Study 1

Leading Life Sciences Data Management Platform

About the Platform

A cloud-based life sciences data management platform built for global pharmaceutical enterprises to streamline multi-country clinical operations. The platform centralizes data collection, regulatory tracking, and analytics across study sites.

Test Value Delivered

- **End-to-End Test Automation:** Developed 420+ automated test scripts across API, UI, and backend layers, increasing regression coverage by **65%** and reducing manual effort by **45%** per release.
- **Compliance-Driven Validation:** Designed 180+ test scenarios aligned with **21 CFR Part 11**, **HIPAA**, and **GxP** frameworks, ensuring audit readiness and traceability.
- **Performance and Data Accuracy Testing:** Simulated concurrent user loads up to **10,000** sessions to validate system stability under production-scale volumes.

Business Impact

- **Faster Release Turnaround:** Achieved **40% reduction** in release validation time, enabling biweekly release cycles instead of monthly.
- **Improved Data Confidence:** Ensured **99.8% data accuracy** in system-to-system synchronization during validation cycles.
- **Simplified Compliance Audits:** Reduced audit preparation time from **3 weeks** to **5 business days**, supported by automated validation documentation.



Case Study 2

Clinical Trial Automation and Monitoring Platform

About the Platform

A digital clinical trial management platform built to help CROs and pharma sponsors manage multi-phase studies, real-time site monitoring, and patient safety reporting across multiple geographies.

Test Value Delivered

- **Automated Regression and Validation:** Implemented automation covering 14 core modules and 75 business-critical workflows, achieving **70% automation coverage** in under 8 weeks.
- **Integration Testing Across Ecosystem:** Verified seamless interoperability among EDC, CTMS, and Safety systems using mock and live API environments – ensuring **100% data consistency** across integrations.
- **Risk-Based Testing Approach:** Adopted a hybrid validation model prioritizing high-risk modules, optimizing QA effort by **35%** without reducing test depth.

Business Impact

- **Accelerated UAT Sign-offs:** Reduced UAT cycle time from 12 days to **7 days**, improving overall release cadence.
- **Zero Critical Defects in Production:** Achieved a defect leakage rate below **0.3%** post-release.
- **Enhanced Sponsor Confidence:** Helped the client secure **three additional clinical partnerships** through improved platform reliability and audit readiness.



Thank You!

